Võ Quang Thành

Junior Full-stack Developer

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EDUCATION

Binh Duong Economics and Technology University — Ho Chi Minh City
Bachelor of Software Engineering — **GPA**: 3.25/4.0

SKILLS

- Programming Languages: C#, Kotlin, JavaScript, TypeScript, SQL, HTML/CSS
- Mobile: Android SDK, Jetpack Compose, MVVM, Coroutines, Room, Firebase (FCM, Auth, Firestore)
- Frameworks & Libraries: Asp.NET Core (Web API/MVC), Node.js (Express.js, NestJS), React.js, RESTful API
- Databases & Tools: SQL Server, PostgreSQL, MySQL, Docker, Redis, Git/GitHub, Postman
- Concepts: OOP, Domain-Driven Design, Clean Architecture, Unit Testing
- Languages: Vietnamese (Native), English (Technical Reading & Documentation)

EXPERIENCE

MID Viet Nam Company Limited — Ho Chi Minh City

Full-stack Developer (Contractor) — 6/2026 – present

• **Insight Center – Enterprise Operations Platform**: Architected and built an end-to-end full-stack operations platform using Node.js (Express.js), TypeScript, Redis, BullMQ, and React 18 under Domain-Driven Design. Engineered hierarchical RBAC, geofenced attendance, and an immutable audit trail with AntV X6 org charts and permission-gated UIs, ensuring sub-500ms API p95 latency and zero data tampering.

• **Automated Payroll, QA Scoring & Alert Engines**: Replaced manual spreadsheets by building an interactive @dnd-kit Kanban board and a 7-status task state machine featuring QA gates, bug-reopen penalties, and optimistic locking. Developed deterministic domain services for statutory payroll computation across a row-locked approval flow with append-only adjustment ledgers and ExcelJS exports. Built an idempotent alert engine using MySQL unique generated columns and batch sweep crons, eliminating duplicate alerts and payroll discrepancies.

ITECCOM Technology Company Limited — Ho Chi Minh City

Fresher Full-stack Developer — 10/2025 – 05/2026

• **Digital Signature Platform:** Designed and implemented a .NET-based digital contract signing platform supporting both single and bulk operations, addressing scalable document processing needs. Developed end-to-end lifecycle management including signer configuration, PDF mapping, dynamic signature placement (image & USB Token), and automated notifications. Architected backend services and optimized SQL Server schemas to handle complex multi-step transactions, reducing manual processing time by 40% and improving system reliability under high-load scenarios.

• **Saigontourist Enterprise CRM:** Enhanced and maintained enterprise CRM systems for loyalty programs and member card management, ensuring reliable customer data operations. Engineered backend services to extract and synchronize near real-time transaction data from the Data Warehouse (DW), reducing data latency and improving data consistency across systems. Optimized

data processing workflows using .NET and SQL Server to support high-volume transactions, ensuring stable performance under concurrent workloads.

PROJECTS

Xiangqi (Chinese Chess) – Android App

Tech Stack: Kotlin, Jetpack Compose, Coroutines, Room, Firebase

- **Github**: github.com/vqt2129a/xiangqi
- **Clean Architecture & DDD:** Structured the app into 6 Gradle modules following the dependency rule. The pure-Kotlin domain has a Game aggregate root that enforces every rule (legal moves, check, checkmate, draws by repetition and move limit) and raises domain events. Use cases depend only on ports (GameRepository, ComputerPlayer), so Room, Firebase and the AI engine are swappable adapters.
- **MVVM & Jetpack Compose:** Built the UI with Jetpack Compose and Navigation Compose, including animated check and result banners. ViewModels expose StateFlow UI state and one-off events through Channels, and are unit-tested on the JVM with fake AI and sound ports.
- **Game AI & Performance:** Implemented an alpha-beta engine (PVS, iterative deepening, Zobrist transposition table, quiescence search) on a compact IntArray board, running on Dispatchers.Default with cooperative cancellation; reaches depth 7 within 3 s. Measured AI think time with Firebase Performance and shrank the release APK with R8.
- **Offline-first & Firebase:** Room is the source of truth and games are restored by replaying moves. A decorator repository backs up finished games to Firestore protected by per-user Security Rules. Anonymous accounts upgrade to Google Sign-In (Credential Manager) with cloud history restore; FCM push notifications deep-link into the unfinished game; Crashlytics for crash reporting.
- **Testing:** 59 unit tests; move generation verified by perft against reference counts (44 / 1,920 / 79,666), plus use-case and ViewModel tests.

Technical Case Study: Enterprise RBAC & Authorization Engine

Tech Stack: Node.js, Redis, BullMQ, Vitest, Docker, MySQL

- **Github**: github.com/vqt2129a/RBAC-Case-Study
- **Architectural Design & Dynamic Data Scoping:** Engineered a Fine-Grained Authorization Engine handling complex access control across multi-level organizational trees (supporting concurrent appointments). Utilized MySQL 8 Recursive CTEs with DAG cycle-detection to dynamically compile hierarchical data scopes (ALL, ORG, OWN) into parameterized SQL predicates.
- **Multi-Tier Caching & Stampede Protection:** Designed and benchmarked a dual-layer caching strategy with Redis MGET by decoupling org subtrees from user auth contexts, slashing authorization latency from 14.2ms to <0.8ms. Implemented an in-memory SingleFlight mechanism to collapse concurrent cache misses, preventing database stampedes during cold starts.
- **Reliable Revocation & Transactional Outbox:** Modeled a 3-tier active cache invalidation pipeline (Synchronous DEL -> BullMQ async retry queue -> Periodic Reconciler) paired with the **Transactional Outbox** pattern, guaranteeing zero-loss cache evictions across role inheritance chains and account lockouts without stale-read vulnerabilities.